
NEURAL-VISUAL DECODING VIA COGNITIVE-GUIDED ADAPTIVE BLURRING AND INFORMATION-CONSTRAINED ALIGNMENT

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ABSTRACT

EEG-based visual decoding aims to establish a mapping between neural signals and visual semantics. However, it remains constrained by the dual challenges of severe information granularity mismatch and the low signal-to-noise ratio (SNR) of EEG signals. Existing approaches typically treat static visual features, ignoring the dynamic selectivity of human vision and the frequency specificity of neural oscillations. To bridge this gap, we propose **CAIA**, a Cognitive-guided Adaptive blurring with Information-Constrained Alignment framework for Neural-Visual decoding. On the visual side, it simulates selective attention to adaptively reduce redundancy. Meanwhile, on the EEG side, it leverages neural oscillation priors and the information bottleneck mechanism to enhance SNR. Specifically, we devise a cognitive-dynamics-based adaptive blurring mechanism that dynamically integrates center-biased and saliency-guided visual cues via cross-modal attention. Furthermore, we introduce a distribution-aware boundary calibration loss to robustly rectify alignment bias caused by outlier samples. Moreover, a cognitively-guided information-screening method is proposed to select task-relevant EEG oscillations. Extensive experiments demonstrate that CAIA improves both subject-dependent and subject-independent average Top-1 and Top-5 accuracy in zero-shot brain-to-image retrieval, significantly outperforming prior methods. Our work validates that optimizing visual information density to match neural granularity offers a more interpretable and robust pathway for neural decoding.

1 Introduction

Understanding the brain’s visual processing mechanism and decoding its neural signals is a cutting-edge research topic bridging neuroscience, cognitive science, and artificial intelligence [1, 2]. Establishing accurate stimulus-activity mappings advances brain-inspired intelligence and supports applications like Brain Computer Interfaces (BCIs) and visual prosthetics [3, 4, 5]. While deep learning has improved neural decoding, EEG’s low SNR and cross-modal granularity mismatch hinder zero-shot generalization [6, 7].

Functional magnetic resonance imaging (fMRI), magnetoencephalography (MEG), and electroencephalography (EEG) constitute the dominant neuroimaging modalities for visual neural decoding [8, 9]. While fMRI excels in spatial resolution, its temporal precision is fundamentally constrained by hemodynamic latency [10]. MEG affords millisecond-scale temporal fidelity but suffers from prohibitive costs and noise-sensitivity [11]. EEG is portable and temporally precise, ideal for real-time applications [12, 13], yet suffers from low SNR due to noise and individual variability [14]. Adaptively extracting visual-related neural features remains a core challenge.

To bridge the representational gap, some studies focus on designing more efficient EEG encoders. For instance, BrainVis [15] hierarchically fuses spatiotemporal features via temporal self-supervised encoding and frequency-domain LSTM. NICE [16] employs a spatial attention module to model inter-electrode correlations. ATM [17] combines spatiotemporal

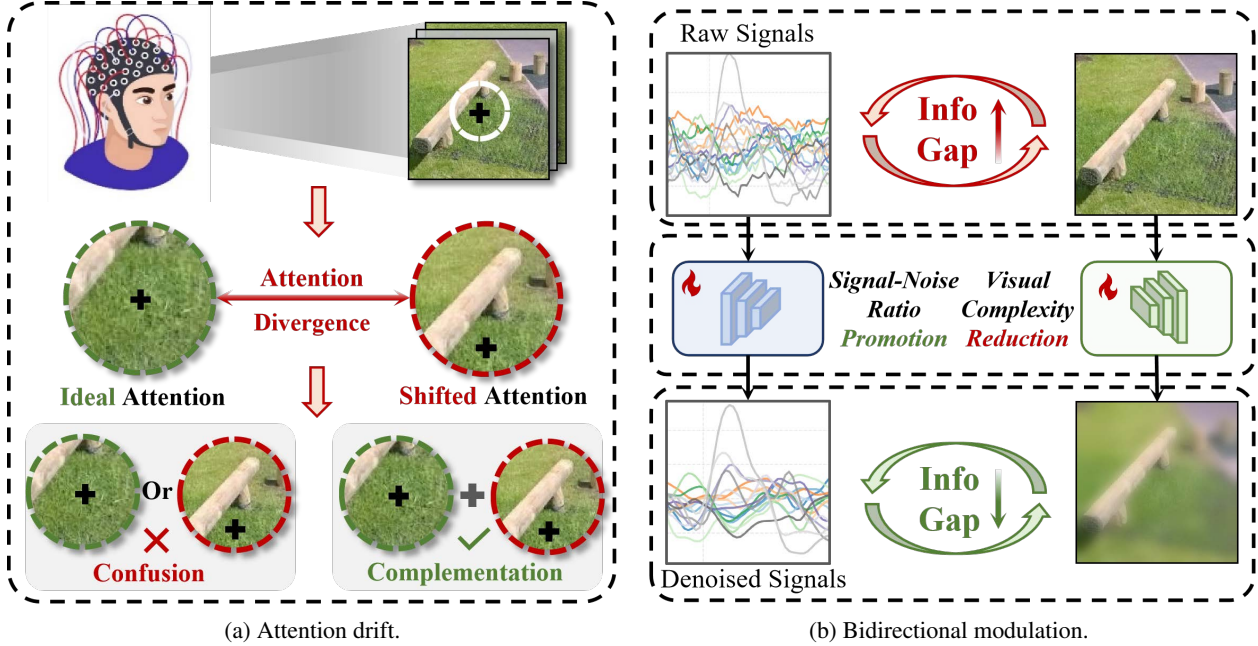


Figure 1: Bidirectional modulation of visual and EEG information densities to bridge the modality gap. (a) Visual information modulation. (b) Integrating center bias and saliency guidance to counteract attention drift.

convolution with channel-wise attention to extract multi-domain EEG features. These methods essentially attempt to distill discriminative patterns from noisy data through complex encoders.

Alternatively, some approaches concentrate on visual feature optimization. Bratrix [18] extracts image background and depth maps to emphasize structural information. While BraVL [6] leverages high-level semantic embeddings from vision-language models to filter pixel-level redundancy. Nevertheless, these methods predominantly optimize unimodal representations without explicitly bridging the representational gap between neural signals and visual stimuli. Recently, UBP [19] proposes blurring images based on the center bias of human visual perception to alleviate this modality gap. However, this approach remains constrained by its rigid center-bias assumption, thereby neglecting attentional shifts triggered by peripheral salient targets. Furthermore, UBP lacks a concomitant mechanism to enhance EEG signal quality, hindering genuine bidirectional optimization and compromising robustness against individual variability and paradigm heterogeneity.

Human visual processing is non-uniform across the visual field. By default, fixations tend toward the image center due to high-resolution foveal vision [20, 21]. However, salient peripheral targets can capture attention during dynamic viewing [22, 23]. As shown in fig. 1a, in typical EEG paradigms such as Rapid Serial Visual Presentation (RSVP), prolonged central fixation may induce attention fatigue and eye drift [24]. Consequently, a static center-blurring strategy could thus suppress neurally encoded peripheral information.

Moreover, EEG signals contain substantial task-irrelevant redundancy [25]. Given the distinct functional roles of neural oscillations across frequency bands [26], such redundancy may reflect a functional mismatch between certain bands and the decoding task. This motivates frequency-band screening—retaining task-relevant bands while suppressing irrelevant ones—to systematically improve SNR. The information bottleneck theory offers a framework that seeks an optimal trade-off between the compression of an input variable and the preservation of relevant information about the downstream task. [27].

To address the above challenges, we propose **CAIA**, a Cognitive-guided Adaptive blurring with Information-Constrained Alignment framework for Neural-Visual decoding. As shown in fig. 1b, CAIA bridges modality gaps via bidirectional information modulation. Visually, a cognitive-dynamics-based blurring mechanism combines center-fixation and saliency-guided paths, dynamically fused via EEG-driven cross-modal attention to simulate selective attention and reduce information density. On the EEG side, a cognitively guided information-screening method integrates neural oscillation priors with the information bottleneck to select task-relevant frequency bands and suppress interference, enhancing SNR. A distribution-aware boundary calibration loss further identifies and calibrates outliers based on

similarity statistics, improving robustness to attention shifts and noise. Our main contributions can be summarized as follows:

- **We propose a cognitive-dynamics-based brain-inspired visual decoding approach** that utilizes center fixation and saliency blurring, dynamically fusing the two visual paths via EEG-driven cross-modal attention to simulate visual selective attention and bridge the modality gap.
- **We introduce a distribution-aware boundary calibration loss** to address outlier samples in EEG-image matching, which dynamically guides them toward the confidence interval for robust cross-modal alignment.
- **We develop a cognitively guided information-screening method** that integrates neuroscientific priors to automatically select task-relevant EEG oscillatory features, suppress irrelevant bands, and enhance SNR.
- **Extensive experiments demonstrate that CAIA surpasses state-of-the-art methods**, achieving +19.4% Top-1 and +16.4% Top-5 accuracy gains over the strongest baseline UBP[19].

2 Related Works

2.1 Visual Neural Decoding

Neural decoding aims to reconstruct perceptual and cognitive content from neural signals, with significant recent progress in motor imagery [28], emotion recognition [29], disease diagnosis [30], and visual decoding [31, 32, 33, 8]. Visual decoding has garnered particular attention for its potential to elucidate the neural mechanisms underlying brain-environment interactions. While fMRI decoding benefits from high spatial resolution for detailed reconstruction [34, 35], EEG decoding capitalizes on millisecond-scale temporal precision, rendering it particularly amenable to real-time brain-computer interfaces and the analysis of dynamic cognitive processes [12, 13].

A core challenge in EEG visual decoding is establishing reliable cross-modal mappings. Early methods used handcrafted features that struggled with non-stationary trial dynamics [36]. Deep learning has since enabled specialized architectures [15, 16, 17, 37, 38, 39, 40] that improve EEG feature encoding. However, EEG’s low SNR inherently limits encoder expressivity and mapping accuracy, without addressing the fundamental inter-modal information granularity mismatch. Recent work has turned to visual feature optimization, via image decomposition or blurring [18, 19, 41]—to mitigate this granularity gap. However, these approaches fail to incorporate a neural feedback mechanism that coordinates dynamic visual adjustment with EEG SNR enhancement, limiting their resilience to shifts in cognitive states and anonymous trials. In contrast, CAIA introduces a bidirectional co-optimization framework that jointly modulates visual and neural representations while explicitly modeling attentional dynamics, thereby achieving more robust cross-modal alignment.

2.2 Multi-Modal Contrastive Learning

Contrastive learning maximizes mutual information between positive pairs and minimizes it for negatives, showing strong results in vision [42] and language [43]. Models like CLIP leverage large-scale image-text pairs for zero-shot generalization [44], inspiring its adaptation to neural decoding by aligning EEG representations with CLIP’s visual space [45]. These methods typically use InfoNCE loss to pull matched pairs together and push unmatched pairs apart in a shared latent space.

Applying this paradigm directly to neural data faces two key issues. First, neural-visual pair quality is lower than natural image-text pairs. EEG contains substantial task-irrelevant noise (e.g., spontaneous oscillations, artifacts), leading to weak positive alignment and spurious correlations [46]. Second, common uniform optimization neglects the inherent similarity distribution in neural decoding, which often approximates a normal distribution [19]. Outliers (e.g., from attention drift or high noise) deviate from this main distribution, biasing gradients. Moreover, increasing model capacity or data augmentation may worsen noise overfitting [47], underscoring the need to integrate cognitive mechanisms—not just data-driven learning. Our CAIA addresses this by incorporating attention drift into a unified loss and employing distribution-aware calibration for cognitively guided contrastive learning, thereby enhancing alignment stability and generalization.

3 Method

3.1 Overall Architecture

CAIA introduces a cognitively guided framework that aligns EEG and visual representations through adaptive blurring and information constraints, as illustrated in fig. 2. During training, EEG features are aligned with blurred image

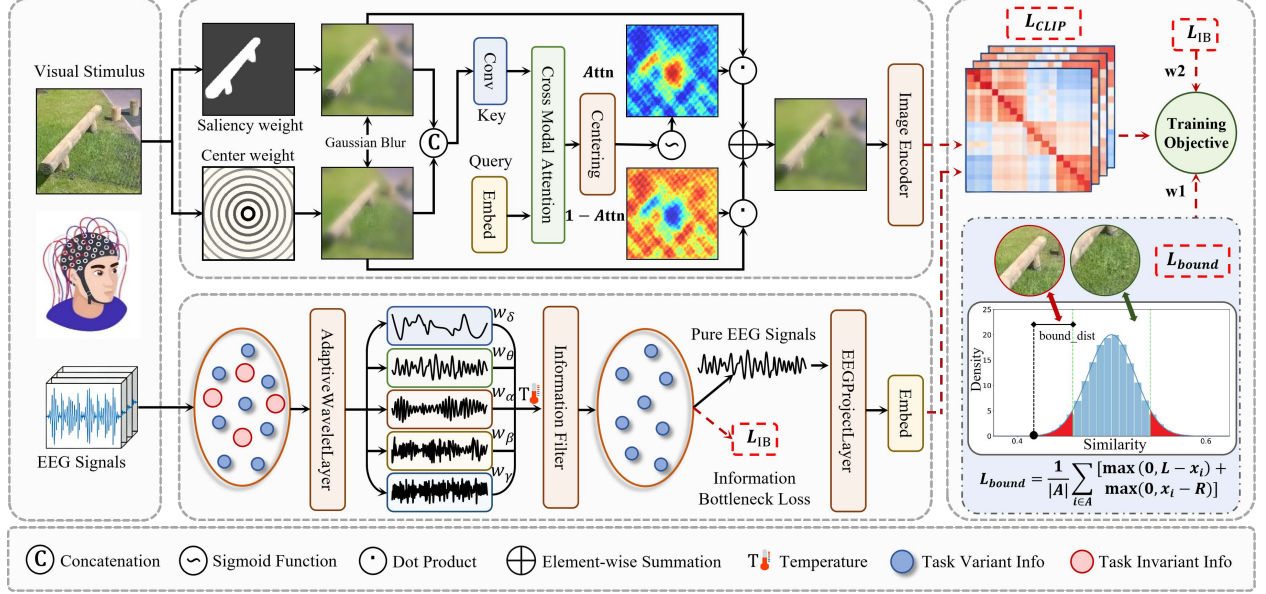


Figure 2: CAIA model framework: (a) dual-path adaptive blurring mechanism. (b) information-bottleneck-guided frequency-selective representation learning. (c) distribution-aware boundary distance loss.

embeddings through contrastive learning, assisted by the boundary loss. During testing, EEG embeddings are matched to visual candidates based on similarity for retrieval evaluation.

3.2 Cognitive-dynamics-based adaptive blurring mechanism

To mitigate information-density mismatch, we propose a dual-path adaptive blurring mechanism. This mechanism is guided by EEG-derived cognitive signals to dynamically select and retain perceptually relevant image regions, moving beyond static blurring strategies. It comprises two parallel blurring paths—saliency-guided differential blurring and center-guided radial blurring, which are adaptively fused via a cross-modal attention module to generate a final blurred image that aligns with the inferred attentional state.

3.2.1 Saliency-Guided Differential Blurring

This branch leverages bottom-up visual saliency [23]. To simulate the “fine perception of salient regions, blurred perception of non-salient regions” characteristic, we apply spatially varying blur intensity inversely proportional to a saliency map. The saliency-guided blurred image X_A is generated as:

$$X_A(i, j) = w_S(i, j) \cdot X(i, j) + [1 - w_S(i, j)] \cdot \mathcal{G}_\sigma(X)(i, j) \quad (1)$$

$$w_S(i, j) = (1 - w_0) \cdot \text{saliency}(i, j) \quad (2)$$

where $w_S(i, j)$ is the saliency-based preservation weight at pixel (i, j) , $w_0 \in [0, 1]$ governs the overall strength of the blurring effect, and $\text{saliency}(i, j)$ is obtained from a salient object detection model, and $\mathcal{G}_\sigma$ is a Gaussian blur operator.

3.2.2 Center-Guided Radial Blurring

This path incorporates the prior of visual center bias—the tendency of the human visual system to direct initial and early fixations toward the geometric center of stimuli [21]. We simulate this characteristic using a radial blur profile that preserves clarity at the center and gradually increases blur toward the periphery [19]. The center-guided blurred image X_B is computed as:

$$X_B(i, j) = w_R(i, j) \cdot X(i, j) + [1 - w_R(i, j)] \cdot \mathcal{G}_\sigma(X)(i, j) \quad (3)$$

$$w_R(i, j) = (1 - w_0) \cdot \exp(-g \cdot d(i, j)) \quad (4)$$

where $w_R(i, j)$ is the center-bias preservation weight at pixel (i, j) , $w_0 \in [0, 1]$ governs the overall strength of the blurring effect, $d(i, j)$ is the normalized distance from the image center, and g controls the blur decay rate.

3.2.3 Cross-Modal Attention Fusion Strategy

The two blurring paths have complementary strengths and context-dependent applicability. To dynamically balance the “default center” and “stimulus capture” mechanisms, we introduce a cross-modal attention fusion strategy. Here, EEG features serve as the query Q , encoding the subject’s cognitive state, while features from the concatenated dual-path blurred images form the key K . This allows the model to estimate pixel-wise fusion weights A that are informed by the neural signal:

$$S = Q^\top K \in \mathbb{R}^{1 \times HW} \quad (5)$$

$$A = \text{sigmoid}(S - \bar{S}) \in \mathbb{R}^{1 \times HW} \quad (6)$$

$$X_{\text{fused}} = A \odot X_A + (1 - A) \odot X_B \quad (7)$$

We adopt the sigmoid activation in lieu of softmax to compute pixel-wise fusion weights. This design avoids the sum-to-one constraint imposed by softmax, which can cause overconfidence in a single region and limit attention allocation across multiple salient areas [48]. The sigmoid function enables independent modulation per pixel, permitting simultaneous attention to several regions. To preserve relative spatial contrast amid this independence, we apply a centering operation by subtracting the mean score $\bar{S}$, thereby amplifying differential attention strength across the image and enhancing the spatial coherence of the fused output.

3.3 Distribution-Aware Boundary Calibration Loss

Prior work observes that the cross-modal similarity between matched EEG-visual pairs tends to follow a Gaussian distribution [19]. Our empirical analysis substantiates this observation (fig. 3).

We regard samples whose similarity lies near the distribution mean as “inliers,” while those deviating significantly—often due to attention drift or high-noise trials—are treated as “outliers.” Such outliers typically reflect substantial cross-modal information mismatch. Optimizing with a uniform contrastive loss risks overfitting to these aberrant samples and may harm generalization.

To mitigate this issue, we propose a distribution-aware boundary distance loss that dynamically identifies outliers and softly draws them toward a plausible similarity range, thereby improving alignment robustness. The loss is defined as:

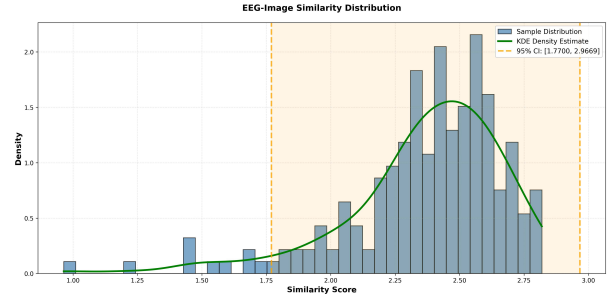


Figure 3: EEG-image similarity distribution.

$$\mathcal{L}_{\text{bound}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} (\ell_i^{\text{left}} + \ell_i^{\text{right}}) \quad (8)$$

$$\ell_i^{\text{left}} = \max(\bar{s} - z_{\alpha/2}\sigma - s_i, 0) \quad (9)$$

$$\ell_i^{\text{right}} = \max(s_i - \bar{s} - z_{\alpha/2}\sigma, 0) \quad (10)$$

Here, s_i denotes the similarity of the i -th matched pair, $\bar{s}$ and σ are the mean and standard deviation of the batch similarities, $z_{\alpha/2}$ is the upper $\alpha/2$ quantile of the standard normal distribution, and $\mathcal{M}$ represents the set of outlier samples falling outside the confidence interval. This soft-pull strategy mitigates outlier-induced overfitting and suppresses interference from attention shifts and transient noise.

3.4 Cognitively guided information-screening method

This module extracts task-relevant visual information from EEG signals by leveraging neuroscientific priors on the frequency-specific functional roles of neural oscillations. Guided by the Information Bottleneck (IB) principle, we learn a compressed representation Z that preserves information about the visual stimulus Y while discarding task-irrelevant variability from the raw EEG signal X :

$$\min_{p(z|x)} I(X; Z) - \beta I(Z; Y) \quad (11)$$

Direct optimization of the mutual information terms is intractable; thus, we adopt a parameterized EEG encoder to approximate Z and employ practical surrogate objectives for the IB formulation.

Maximizing $I(Z; Y)$: The contrastive loss aligns Z with the visual embedding Y , effectively maximizing a lower bound on $I(Z; Y)$ to promote task-relevant information.

Minimizing $I(X; Z)$: Since $I(X; Z) = H(Z) - H(Z | X) \leq H(Z)$ and $H(Z | X) = 0$ for a deterministic encoder, minimizing $H(Z)$ serves as a surrogate objective for reducing $I(X; Z)$, where $H(\cdot)$ is entropy.

To instantiate IB-driven filtering, we first decompose the raw EEG into sub-bands via an adaptive wavelet transform, where the decomposition scale $k_i = k_i^0 \cdot \gamma_i$ is made learnable through parameter γ_i , enabling task-specific spectral boundary tuning. Subsequently, an information bottleneck layer learns soft selection weights $\{m_i\}$ over these bands using a temperature-controlled softmax.

$$m_i = \frac{\exp(w_i/\tau)}{\sum_{j=1}^n \exp(w_j/\tau)} \quad (12)$$

where w_i denote learnable parameters and τ governs distribution sparsity. The denoised representation Z is obtained via weighted fusion of sub-band features using $\{m_i\}$. Consequently, the entropy of the selection distribution $H(M)$ approximates $H(Z)$.

$$H(M) = - \sum_{i=1}^n m_i \log m_i \quad (13)$$

Combining the components, our full training objective is:

$$\mathcal{L}_{\text{Overall}} = \underbrace{\mathcal{L}_{\text{CLIP}}}_{\text{maximize } I(Y; Z)} + w_1 \cdot \mathcal{L}_{\text{bound}} + w_2 \cdot \underbrace{\left(- \sum_{i=1}^n m_i \log m_i \right)}_{\text{minimize } I(X; Z) \approx H(M)} \quad (14)$$

Where $\mathcal{L}_{\text{CLIP}}$ is the contrastive loss based on CLIP [44].

4 Experiment

4.1 Datasets and Implementation Details

We evaluate the proposed framework and comparison methods on two publicly available neuroimaging datasets derived from the THINGS stimulus library.

THINGS-EEG [14] records electroencephalography (EEG) from 10 subjects under a RSVP paradigm. The training set comprises 1,654 categories (10 images per category, 4 repetitions each), while the test set contains 200 novel categories (1 image per category, 80 repetitions). Following UBP [19], we enhance the signal-to-noise ratio by averaging repeated trials of the same stimulus, using a post-stimulus window of 0–250 ms and selecting 17 visually related channels spanning occipital, parietal, and lateral temporal regions.

THINGS-MEG [49] contains magnetoencephalography (MEG) recordings from 4 subjects (271 channels). The training set covers all 1,654 categories (12 images per category, single presentation), and the test set comprises 200 novel categories (1 image per category, 12 repetitions).

We adopt the ResNet-50 vision encoder from the OpenCLIP library as our visual feature extractor. All experiments were implemented in PyTorch on an NVIDIA RTX 3090 GPU. We used the Adam optimizer with a batch size of 128 and a learning rate of 1e-4 for subject-dependent settings, and a batch size of 256 with a learning rate of 1e-5 for subject-independent settings. The loss weights w_1 and w_2 in eq. (14) are set to 0.01. The blur decay rate g is set to 3.

4.2 Overall Performance

We compare the proposed method with state-of-the-art approaches, including BraVL [6], NICE [16], ATM [17], CogCap [41], VE-SDN [45], MB2C [50], Neural-MCRL [51], and UBP [19].

As summarized in table 1 and table 2, CAIA establishes a new state of the art on both EEG and MEG test sets under subject-dependent and subject-independent protocols.

On THINGS-EEG, CAIA attains subject-dependent Top-1/Top-5 accuracy of **70.3% / 96.1%**, surpassing the strong UBP baseline by +19.4% / +16.4%. This marked improvement validates that the proposed adaptive blurring mechanism and EEG-side information screening collectively mitigate the cross-modal granularity mismatch. In the more challenging subject-independent setting, CAIA achieves **16.4% / 45.7%** (UBP: 12.4% / 33.4%), demonstrating enhanced robustness to inter-subject variability through the distribution-aware boundary loss.

Table 1: Overall accuracy (%) of 200-way zero-shot retrieval on THINGS-EEG: Top-1 and Top-5.

Method	Subject 1		Subject 2		Subject 3		Subject 4		Subject 5		Subject 6		Subject 7		Subject 8		Subject 9		Subject 10		Avg	
	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5
Subject-dependent (intra-subject): train and test on one subject																						
BraVL [6]	6.1	17.9	4.9	14.9	5.6	17.4	5.0	15.1	4.0	13.4	6.0	18.2	6.5	20.4	8.8	23.7	4.3	14.0	7.0	19.7	5.8	17.5
NICE [16]	13.2	39.5	13.5	40.3	14.5	42.7	20.6	52.7	10.1	31.5	16.5	44.0	17.0	42.1	22.9	56.1	15.4	41.6	17.4	45.8	16.1	43.6
ATM [17]	25.6	60.4	22.0	54.5	25.0	62.4	31.4	60.9	12.9	43.0	21.3	51.1	30.5	61.5	38.8	72.0	34.4	51.5	29.1	63.5	27.1	58.1
CogCap [41]	27.2	59.5	28.7	57.0	37.2	66.1	37.7	63.2	21.8	47.8	31.6	58.1	32.8	59.6	47.6	73.5	33.4	57.7	35.1	63.6	33.3	60.6
VE-SDN [45]	32.6	63.7	34.4	69.9	38.7	73.5	39.8	72.0	29.4	58.6	34.5	68.8	34.5	68.3	49.3	79.8	39.0	69.6	39.8	75.3	37.2	70.0
MB2C [50]	23.6	56.3	22.6	50.5	26.3	60.1	34.8	67.0	21.3	53.0	31.0	62.3	25.0	54.8	39.0	69.3	27.5	59.3	33.1	70.8	28.4	60.3
Neural-MCRL [51]	27.5	64.0	28.5	61.5	37.0	69.0	35.0	66.0	22.5	51.5	31.5	61.0	31.5	62.5	42.0	74.5	30.5	59.5	37.5	71.0	32.4	64.1
UBP [19]	41.2	70.5	51.2	80.9	51.2	82.0	51.1	76.9	42.2	72.8	57.5	83.5	49.0	79.9	58.6	85.8	45.1	76.2	61.5	88.2	50.9	79.7
CAIA (Ours)	74.0	98.0	80.0	97.5	69.0	94.0	51.0	92.5	67.5	94.0	67.5	96.5	63.0	95.5	83.0	99.0	69.0	96.5	79.0	97.5	70.3	96.1
Subject-independent (inter-subject): leave one subject out for test																						
BraVL [6]	2.3	8.0	1.5	6.3	1.4	5.9	1.7	6.7	1.5	5.6	1.8	7.2	2.1	8.1	2.2	7.6	1.6	6.4	2.3	8.5	1.8	7.0
NICE [16]	7.6	22.8	5.9	20.5	6.0	22.3	6.3	20.7	4.4	18.3	5.6	22.2	5.6	19.7	6.3	22.0	5.7	17.6	8.4	28.3	6.2	21.4
ATM [17]	10.5	26.8	7.1	24.8	11.9	33.8	14.7	39.4	7.0	23.9	11.1	35.8	16.1	43.5	15.0	40.3	4.9	22.7	20.5	46.5	11.9	33.8
CogCap [41]	16.3	42.3	16.2	37.9	8.8	26.8	15.4	37.6	10.1	31.7	14.0	35.4	10.7	26.9	13.9	34.2	9.0	32.4	15.3	38.6	13.0	34.4
MB2C [50]	10.5	28.1	11.3	32.8	8.8	27.6	13.6	33.5	10.6	27.5	12.1	33.1	11.5	31.8	12.0	32.1	12.1	31.3	16.1	42.1	11.9	32.0
Neural-MCRL [51]	13.0	31.5	12.0	30.5	14.5	35.5	12.5	35.5	11.5	29.0	13.5	35.5	14.0	36.0	18.5	38.5	13.5	32.5	17.0	39.0	14.0	34.3
UBP [19]	11.5	29.7	15.5	40.0	9.8	27.0	13.0	32.3	8.8	33.8	11.7	31.0	10.2	23.8	12.2	32.2	15.5	40.5	16.0	43.5	12.4	33.4
CAIA (Ours)	14.0	39.5	21.5	57.5	13.5	37.0	14.0	53.5	12.0	40.5	16.0	43.0	16.5	42.0	15.0	40.5	10.0	35.5	31.0	67.5	16.4	45.7

On THINGS-MEG, CAIA achieves subject-dependent accuracies of **31.8%** / **62.6%** and subject-independent accuracies of **7.5%** / **21.8%**, outperforming UBP by +5.1% / +7.4% and +5.3% / +11.4%, respectively. These results indicate that the framework can leverage neural oscillation priors to identify stable cross-modal relationships even with weaker inter-subject signals, evidencing promising cross-modal generalization.

4.3 Ablation Experiments

We conduct ablation studies to evaluate three core components of CAIA in isolation: the adaptive blurring mechanism (*ATTN*), the information-screening representation learner (*IBWave*), and the boundary calibration loss (*L_{bound}*). Results are reported in table 3.

Disabling all modules (leaving only the EEG projection head, center-biased blurring, and the CLIP contrastive loss) yields a baseline performance of 54.5% Top-1 and 87.3% Top-5 accuracy on THINGS-EEG. Enabling *ATTN* alone improves performance to 62.1% / 92.6%, demonstrating the efficacy of center-saliency fusion in reducing visual redundancy. Using *IBWave* in isolation achieves 64.0% / 91.6%, indicating that frequency-band screening suppresses EEG noise and enhances the SNR. Employing *L_{bound}* alone leads to a slight degradation (52.4% / 85.7%). This is expected, as it serves as an anomaly-identification mechanism rather than a correction strategy. Its role is to detect outlying cross-modal pairs, creating the precondition for any subsequent adjustment method to take effect. As a decoupled module, it can be flexibly combined with other techniques that adjust anomalous representations.

Pairwise combinations reveal synergistic effects. *ATTN* + *IBWave* attains 64.9% / 93.6%, highlighting complementary benefits from visual and EEG-side optimization. *ATTN* + *L_{bound}* yields 64.0% / 93.0%, verifying that the boundary loss can rectify attention-driven outliers post-blurring. *IBWave* + *L_{bound}* achieves 64.9% / 92.4%, as noise-suppressed EEG features facilitate more reliable distribution modeling.

The full CAIA model integrates all three components, achieving the best overall performance of 70.3% Top-1 and 96.1% Top-5 accuracy. This optimal result demonstrates the effectiveness of the complete design loop: *IBWave* supplies denoised EEG features, *ATTN* generates cogni-

Table 2: Overall Top-1/5 accuracy (%) on THINGS-MEG for 200-way zero-shot recognition.

Method	Sub 1		Sub 2		Sub 3		Sub 4		Avg	
	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5
Subject-dependent										
MB2C [50]	9.3	33.6	20.6	49.2	18.2	44.3	10.2	33.6	14.6	39.9
NICE [16]	11.5	35.6	25.7	54.4	21.0	47.8	11.2	35.2	17.4	43.3
ATM [17]	8.0	29.3	30.2	61.5	20.3	50.5	11.8	33.3	17.6	43.7
UBP [19]	<u>15.0</u>	<u>38.0</u>	<u>46.0</u>	<u>80.5</u>	<u>27.3</u>	<u>59.0</u>	<u>18.5</u>	<u>43.5</u>	<u>26.7</u>	<u>55.2</u>
CAIA	18.5	38.0	47.0	89.0	37.0	73.5	24.5	50.0	31.8	62.6
Subject-independent										
UBP [19]	2.0	5.7	1.5	17.2	2.7	10.5	2.5	8.0	2.2	10.4
CAIA	9.5	27.0	4.0	15.0	9.5	21.5	7.0	23.5	7.5	21.8

Table 3: Ablation study on THINGS-EEG (average Top-1 / Top-5, %).

Modules			Avg	
IBWave	ATTN	<i>L_{bound}</i>	Top-1	Top-5
×	×	×	54.5	87.3
×	×	✓	52.4	85.7
×	✓	×	62.1	92.6
×	✓	✓	64.0	93.0
✓	×	×	64.0	91.6
✓	×	✓	64.9	92.4
✓	✓	×	64.9	93.6
✓	✓	✓	70.3	96.1

tively aligned blurred images, and L_{bound} calibrates residual mismatches, collectively establishing robust cross-modal alignment.

4.4 Model Analysis

4.4.1 Cross-Modal Attention Analysis

As shown in fig. 4, the attention heatmap inferred by the adaptive blurring mechanism aligns with image semantics, avoiding uniform blurring or overreliance on a single pathway. This confirms that the cross-modal attention module can precisely identify regions where visual detail should be preserved.

To evaluate the effectiveness of the cross-modal attention module, we conduct visualization with several alternative image-processing strategies, including the original image, center-only blur, saliency-only blur, fixed linear blending, and a global learnable mask. Quantitative results in table 4 show that static or fixed fusion rules fail to adapt to dynamic attention shifts across experiments, whereas the EEG-guided attention mechanism dynamically modulates fusion weights, balancing center-bias and saliency cues to better approximate human selective attention.

4.4.2 Information-Guided Frequency Band Screening Analysis

After information-guided screening, the Beta band receives the highest selection weight (See Appendix A.4 for details), which aligns with its well-established function in active visual attention, feature integration, and target recognition. Beta oscillations are implicated in mediating information flow between visual and frontoparietal attention networks, thereby facilitating attentional focus [52, 53].

Time-frequency visualizations further demonstrate the effectiveness of signal denoising via the proposed information-bottleneck-guided screening module (fig. 5). The signal amplitude is compressed from approximately $[-1, 1]$ to $[-0.3, 0.3]$, indicating the suppression of task-irrelevant variability. High-frequency components above 40 Hz—often associated with EMG artifacts—are markedly attenuated, confirming effective filtering of Gamma-band interference. Suppression below 5 Hz in certain channels further indicates the removal of non-task-related low-frequency activity.

The EEG topography (fig. 5) exhibits localized signal intensity over a confined occipital-parietal region, which is consistent with prior reports [25]. This spatial pattern corresponds to the functional anatomy of the primary visual cortex (V1–V4) and attentional modulation zones along the intraparietal sulcus, which are known to be prominently activated during object recognition.

4.4.3 Calibration

Effect of Distribution-Aware Boundary Loss

As depicted in fig. 6, the distribution-aware boundary loss effectively calibrates the EEG-image similarity distribution. Under such constraint, the distribution exhibits increased compactness, a higher mean similarity, and fewer outliers beyond the confidence region. These results validate that the proposed loss mitigates alignment biases caused by attention shifts or noisy trials, thereby enhancing the robustness of cross-modal matching.

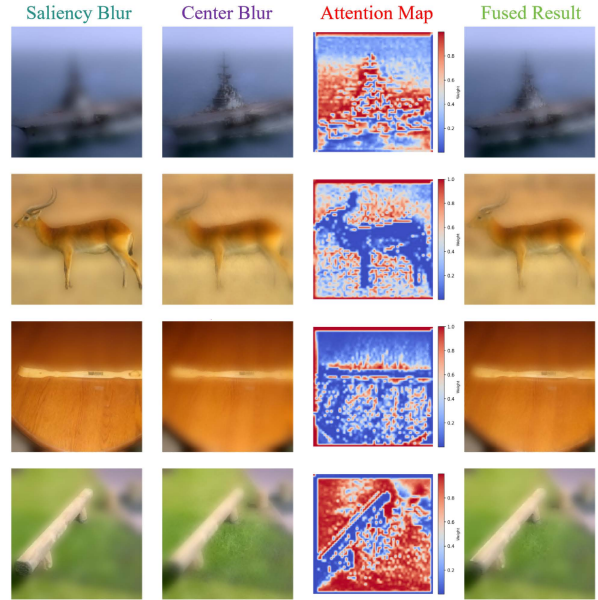


Figure 4: Visualization of the cognitive-dynamics-based adaptive blurring mechanism. The red region in the attention heatmap is biased toward center bias, and the blue region is biased toward saliency guidance.

Table 4: Image-fusion comparison on THINGS-EEG (average Top-1 / Top-5, %).

Method	Top-1	Top-5
Original	64.6	92.1
Center_blur	64.9	92.4
Saliency_blur	59.3	88.0
Linear	63.2	91.5
Learnable_mask	64.1	92.1
Attention (Ours)	70.3	96.1

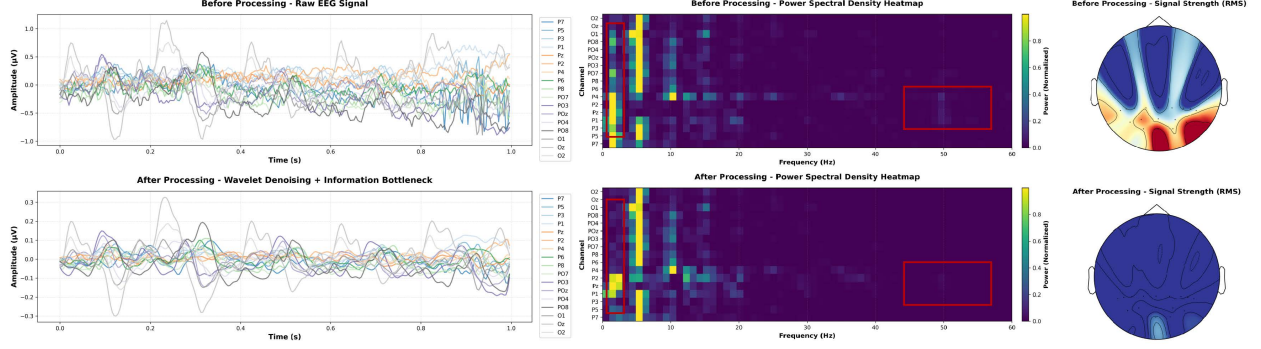


Figure 5: Visualization of EEG signals before and after information-guided frequency band screening in the time domain (left), frequency domain (middle), and EEG topographic map (right).

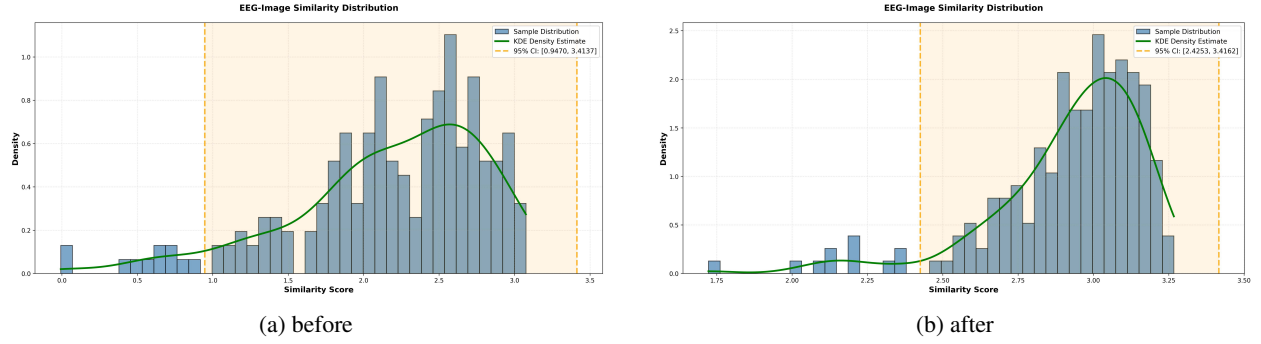


Figure 6: EEG-image similarity distribution before and after calibration using the distribution-aware loss.

5 Conclusion

CAIA presents a cognitively guided framework that mitigates the information granularity mismatch in EEG-visual decoding via bidirectional modulation. Visually, an adaptive blurring mechanism fuses center-biased and saliency-guided cues through EEG-driven cross-modal attention. On the neural side, an information-bottleneck-guided encoder selectively enhances task-relevant frequency bands to improve SNR, while a distribution-aware boundary loss calibrates alignment outliers.

Extensive experiments on THINGS-EEG and THINGS-MEG demonstrate that CAIA consistently surpasses state-of-the-art methods in both subject-dependent and subject-independent zero-shot retrieval. Ablation studies validate each component’s contribution, and visual analyses confirm effective modeling of attentional selectivity and spectral noise suppression.

A limitation is that the adopted cognitive priors—center bias and saliency capture—presuppose an experimental paradigm with sustained central fixation. Their effectiveness may degrade under free-viewing conditions or paradigms without a well-defined fixation point. This work offers a principled, cognition-informed approach to EEG-visual alignment, advancing both performance and interpretability. Future work may extend the framework to multi-subject modeling and real-time BCI scenarios.

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6 A. Supplementary Material

6.1 Details of the Datasets

Things-EEG [14] is a large-scale neuroimaging resource designed to facilitate zero-shot visual object recognition from neural signals. The dataset comprises electroencephalography recordings from ten healthy adult participants with normal or corrected-to-normal vision. EEG signals were acquired using a 64-channel Brain Vision ActiChamp cap arranged according to the extended 10–20 system, sampled at 1,000 Hz with online band-pass filtering (0.1–100 Hz) and referenced to the average of all channels. Visual stimulation followed a Rapid Serial Visual Presentation (RSVP) protocol. An orthogonal detection task was employed to ensure sustained attention. Each participant completed multiple experimental sessions, yielding a total of 82,160 trials per subject. The stimulus set was drawn from the THINGS image database, encompassing 1,854 distinct object concepts organized into 27 higher-level categories (e.g., animals, vehicles, food items, tools, sports equipment). These concepts were partitioned into 1,654 training categories (10 images per category, 4 repetitions each) and 200 hold-out test categories (1 image per category, 80 repetitions for signal averaging).

Things-MEG [49] contains magnetoencephalography recordings from 4 healthy participants acquired using a 271-channel whole-head system. The experimental protocol involved 500 ms stimulus presentation followed by a $1,000 \pm 200$ ms blank interval, with participants performing an oddball detection task. The dataset comprises 1,654 training concepts (12 images per category, single presentation) and 200 test concepts (1 image per category, 12 repetitions). MEG data were band-pass filtered (0.1–100 Hz), epoched from 0–250 ms post-stimulus, and downsampled to 200 Hz, with trial averaging applied to enhance sensor-level signal stability.

6.2 Comparison Methods

BraVL [6] develops a multimodal variational autoencoding framework based on mixture-of-product-of-experts to jointly embed brain activity, visual features, and textual descriptions into a shared latent representation. It maximizes both intra- and inter-modality mutual information to ensure consistent joint generation and improve data efficiency in zero-shot neural decoding scenarios.

NICE [16] presents a self-supervised framework that aligns image and EEG representations via contrastive learning for zero-shot object recognition. It maximizes the cosine similarity between paired image stimuli and EEG responses while pushing non-paired samples apart, enabling the EEG encoder to extract semantics consistent with visual features without categorical labels. Additionally, plug-and-play self-attention and graph attention modules are introduced to capture spatial correlations among EEG channels, providing interpretable evidence of localized brain activity during visual processing.

ATM [17] proposes an end-to-end zero-shot visual reconstruction framework that projects EEG signals into a shared embedding space aligned with CLIP. It employs channel-wise Transformer encoders with temporal-spatial convolution to capture semantic and spatial correlations, facilitating tasks like classification, retrieval, and image generation.

CogCap [41] presents a unified multimodal decoding framework that incorporates text and depth alongside EEG and image modalities. It uses dedicated encoders to align EEG with each modality, capturing “beyond-image” information, and a lightweight diffusion prior to map EEG embeddings into the CLIP space for high-fidelity reconstruction.

VE-SDE [45] introduces a framework that disentangles semantic-related and domain-specific features via mutual information maximization-minimization adversarial learning. It constructs a joint semantic space to reproject visual and EEG embeddings, mitigating the modality gap, and employs an intra-class geometric consistency constraint to maintain uniform distance gaps within categories.

MB2C [50] employs a dual-GAN architecture with cycle consistency loss and contrastive learning to generate and inversely translate modality-related features. This approach narrows the modality gap between EEG and visual features, ensuring semantically aligned embeddings reside in a shared representation space.

Neural-MCRL [51] achieves multimodal alignment via semantic bridging and cross-attention mechanisms, ensuring intra-modal semantic completion and inter-modal consistency. It introduces NESTA to adaptively capture spectral-temporal EEG patterns with subject-specific transformations for robust visual decoding.

UBP [19] proposes an Uncertainty-aware Blur Prior framework that quantifies uncertainty within paired brain-visual data to estimate the inherent mismatch. It dynamically blurs high-frequency details of visual stimuli accordingly, thereby reducing information discrepancy and enhancing alignment robustness for zero-shot neural decoding.

6.3 Additional Experimental Results

This section provides extended experimental results and analyses.

6.3.1 Saliency Detection Model Comparison

Table 5: Comparison of different saliency detection models within CAIA on THINGS-EEG (200-way zero-shot retrieval, subject-dependent, Top-1 / Top-5 %).

Method	Sub 1		Sub 2		Sub 3		Sub 4		Sub 5		Sub 6		Sub 7		Sub 8		Sub 9		Sub 10		Avg	
	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5
U ² Net	67.5	95.5	75.0	96.5	73.0	96.5	55.5	91.0	63.0	92.0	75.0	96.5	62.0	94.5	80.5	97.5	63.0	93.0	77.5	97.0	69.2	95.0
Spectral Residual	62.0	91.0	71.5	96.0	71.0	96.0	47.5	88.0	66.5	95.0	71.0	96.5	62.0	94.0	78.0	97.5	62.0	94.0	74.5	96.5	66.6	94.5
Fine-Grained	74.0	98.0	80.0	97.5	69.0	94.0	51.0	92.5	67.5	94.0	67.5	96.5	63.0	95.5	83.0	99.0	69.0	96.5	79.0	97.5	70.3	96.1

We evaluate three saliency detection backends that can be plugged into our saliency-guided blurring branch (table 5). U²Net is a deep salient object detector pre-trained on generic saliency benchmarks. Spectral Residual and Fine-Grained are both from the OpenCV library, the former identifies salient regions via spectral irregularities, while the latter computes saliency based on color and spatial contrast, naturally emphasizing visually meaningful structures.

Fine-Grained achieves the highest accuracy (70.3% Top-1, 96.1% Top-5). We attribute this to its better alignment with human visual mechanisms. Its contrast-based saliency closely mimics how the early visual cortex differentially weights edges and homogeneous regions, guiding the adaptive blurring to preserve perceptually critical information while suppressing irrelevant background texture. One possible reason for the lower performance of U²Net is that, as a generic pre-trained model, its output saliency distribution may not perfectly match the image characteristics of the THINGS-EEG dataset, leading to sub-optimal blur masks for this specific decoding task. Spectral Residual, being sensitive to frequency outliers, sometimes amplifies textural noise rather than consistent object structures, introducing incorrect preservation priorities.

6.3.2 Frequency Filter Comparison

Table 6: Comparison of our learnable information-guided screening (IBWave) with generic band-pass filters on THINGS-EEG (200-way zero-shot retrieval, subject-dependent, Top-1 / Top-5 %).

Method	Sub 1		Sub 2		Sub 3		Sub 4		Sub 5		Sub 6		Sub 7		Sub 8		Sub 9		Sub 10		Avg	
	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5
High-Frequency	22.0	46.5	24.5	63.0	11.5	43.0	18.5	48.5	9.0	29.5	20.5	53.0	23.0	57.0	24.5	58.5	23.0	56.5	27.5	58.0	20.4	51.4
Low-Frequency	43.5	86.5	38.5	78.5	48.0	88.5	39.0	82.5	44.5	84.0	44.5	87.5	52.0	87.0	61.5	91.5	40.0	84.0	53.5	95.5	46.5	86.6
Mid-Frequency	62.0	93.5	63.5	93.0	65.5	94.0	52.5	86.5	59.0	86.5	66.0	94.5	44.5	84.0	51.5	89.5	45.5	84.0	70.5	95.0	58.1	90.1
IBWave (Ours)	74.0	98.0	80.0	97.5	69.0	94.0	51.0	92.5	67.5	94.0	67.5	96.5	63.0	95.5	83.0	99.0	69.0	96.5	79.0	97.5	70.3	96.1

To verify that the gains of our cognitive-guided information-screening module stem from task-adaptive band selection rather than mere frequency-domain denoising, we compare IBWave against three static band-pass filters (table 6): a high-frequency filter ($> 40\text{Hz}$), a low-frequency filter ($< 8\text{Hz}$), and a mid-frequency filter (8 - 40Hz).

Among the generic filters, the mid-frequency filter achieves the best performance (58.1% Top-1, 90.1% Top-5), consistent with the Beta-dominant weighting learned by our model and the established role of Beta oscillations in active visual attention. However, its accuracy remains substantially lower than that of IBWave (70.3% Top-1, 96.1% Top-5). The key difference is that static filters rely on fixed, hand-crafted frequency boundaries, whereas IBWave dynamically optimizes both the decomposition scales and band-selection weights in a task-driven manner. This allows the model to adaptively retain task-relevant spectral components and suppress interference that would otherwise pass through a rigid band-pass filter. The results confirm that the observed gains are attributable to learnable, task-adaptive frequency screening rather than generic denoising.

6.3.3 Blur Strength and Confidence Level Sensitivity

Both excessively light ($w_0 = 0.3$) and heavy ($w_0 = 0.7$) blurring degrade performance (table 7), indicating that a moderate w_0 strikes the necessary balance between information reduction and semantic preservation.

Performance remains stable across $\alpha \in \{90\%, 95\%, 99\%\}$ (table 8), demonstrating that the boundary calibration loss is robust to the choice of confidence threshold.

Table 7: Sensitivity analysis of the blur strength parameter w_0 on THINGS-EEG (200-way zero-shot retrieval, subject-dependent, Top-1 / Top-5 %).

w_0	Sub 1		Sub 2		Sub 3		Sub 4		Sub 5		Sub 6		Sub 7		Sub 8		Sub 9		Sub 10		Avg	
	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5
Light ($w_0 = 0.3$)	68.0	93.0	77.0	94.0	82.0	99.0	50.0	97.0	60.5	92.5	80.5	97.5	58.0	93.0	85.0	98.5	57.5	94.5	76.0	97.5	69.5	95.7
Heavy ($w_0 = 0.7$)	68.0	95.0	77.0	97.0	69.0	92.5	58.0	95.0	59.0	92.0	70.5	95.5	57.0	92.0	74.5	95.5	58.5	92.0	77.0	94.5	66.9	94.1
Standard ($w_0 = 0.5$)	74.0	98.0	80.0	97.5	69.0	94.0	51.0	92.5	67.5	94.0	67.5	96.5	63.0	95.5	83.0	99.0	69.0	96.5	79.0	97.5	70.3	96.1

Table 8: Sensitivity analysis of the confidence level α in the boundary calibration loss on THINGS-EEG (200-way zero-shot retrieval, subject-dependent, Top-1 / Top-5 %).

α	Sub 1		Sub 2		Sub 3		Sub 4		Sub 5		Sub 6		Sub 7		Sub 8		Sub 9		Sub 10		Avg	
	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5	T1	T5
99%	63.0	97.0	80.0	97.5	79.0	99.0	50.5	88.5	63.0	93.0	77.5	95.0	64.5	93.5	80.0	97.0	64.5	94.0	80.0	97.5	70.2	95.2
90%	69.5	97.5	78.5	96.5	69.0	97.5	56.0	92.5	58.5	90.5	79.0	95.5	64.0	93.0	80.0	97.5	63.5	90.0	80.5	96.0	69.9	94.7
95%	74.0	98.0	80.0	97.5	69.0	94.0	51.0	92.5	67.5	94.0	67.5	96.5	63.0	95.5	83.0	99.0	69.0	96.5	79.0	97.5	70.3	96.1

6.3.4 Computational Overhead Analysis

We compare the computational cost of CAIA and UBP [19] in Table 9. Although CAIA introduces additional parameters and FLOPs due to the dual-path blurring and information-screening modules, its inference latency remains well within the practical constraints of real-time BCI systems. Prior work indicates that EEG decoding latency should be controlled below 200 ms [54], and even stricter limits for implantable devices (~ 50 ms) are relaxed for non-invasive EEG [55]. Furthermore, the wireless transmission delay of commercial EEG acquisition devices alone can reach tens of milliseconds [56]. Within this context, CAIA’s inference latency (14.29 ms) constitutes only a modest increase over UBP (5.88 ms) and falls entirely within the tolerable latency budget. The increased training cost is offset by the substantial performance gain.

Table 9: Computational overhead comparison. Inference latency and training step time are measured on an NVIDIA RTX 3090.

Method	Params (M)	FLOPs (G)	Inference (ms)	Train Step (ms)
UBP	43.72	5.423	5.88 ± 0.58	8.35 ± 0.49
CAIA	107.69	6.559	14.29 ± 0.44	34.79 ± 0.42

6.3.5 Full Ablation and Image-Fusion Tables

For completeness, table 10 lists the full ablation results across all subjects, and table 11 provides the detailed comparison of different image-fusion strategies. These tables expand the summary statistics given in the main paper and support the conclusions drawn in the main text.

6.4 Extended Visualizations

This section supplements the main paper with additional qualitative and visualization analyses.

fig. 7 shows representative cases where the Top-1 prediction is correct. The retrieved images align with the target stimuli at the level of visual category, validating that the model has learned discriminative cross-modal mappings.

fig. 8 presents failure cases for further inspection. Although the Top-1 retrieval is incorrect, semantically plausible or visually similar alternatives frequently appear among the Top-5 results. This pattern suggests that the model reliably captures coarse-grained visual and semantic information from EEG signals, while fine-grained discrimination between visually or semantically similar categories remains a challenge. Together, these qualitative results corroborate the robustness of the learned representations.

fig. 9 offers a low-dimensional visualization of the learned latent spaces using UMAP. The two distributions (figs. 9a and 9c) exhibit striking topological similarity, confirming that the alignment mechanism successfully brings neural and visual representations into a shared geometric structure. fig. 9b provides a joint UMAP plot where green points denote

Table 10: Ablation study results (%) of 200-way zero-shot retrieval on THINGS-EEG: Top-1 and Top-5.

Modules			Subject 1		Subject 2		Subject 3		Subject 4		Subject 5		Subject 6		Subject 7		Subject 8		Subject 9		Subject 10		Avg	
IBWave	ATTN	L_{bound}	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5
x	x	x	51.5	82.0	55.0	89.0	58.0	89.5	45.0	82.0	45.5	79.0	57.5	92.0	59.0	91.5	64.5	92.5	47.5	83.0	61.5	92.5	54.5	87.3
x	x	✓	51.5	85.0	43.5	80.5	49.5	84.5	48.0	83.0	42.0	81.0	56.5	90.5	59.5	87.5	62.5	93.0	42.0	80.0	68.5	91.5	52.4	85.7
x	✓	x	60.5	91.0	66.0	94.5	61.5	93.5	54.0	90.5	52.0	84.0	68.5	95.5	60.5	93.0	67.5	95.5	58.5	91.5	71.5	96.5	62.1	92.6
x	✓	✓	66.5	92.5	70.0	94.5	66.5	95.0	54.5	89.5	52.0	87.0	69.5	95.5	61.5	91.0	69.5	95.5	59.0	92.0	70.5	97.0	64.0	93.0
✓	x	x	66.5	93.5	63.0	93.5	61.0	90.5	49.0	84.5	59.5	86.5	68.0	95.0	65.5	92.5	75.5	98.0	61.0	88.5	70.5	93.5	64.0	91.6
✓	x	✓	68.0	95.5	69.0	93.5	62.0	90.5	51.0	85.0	62.0	89.0	69.5	95.5	61.5	91.5	74.5	97.5	60.0	90.5	71.0	95.0	64.9	92.4
✓	✓	x	66.5	94.5	68.5	95.0	66.5	94.0	53.5	89.0	57.0	90.0	70.0	95.5	61.0	92.0	73.5	98.5	60.0	92.5	72.5	95.0	64.9	93.6
✓	✓	✓	74.0	98.0	80.0	97.5	69.0	94.0	51.0	92.5	67.5	94.0	67.5	96.5	63.0	95.5	83.0	99.0	69.0	96.5	79.0	97.5	70.3	96.1

Table 11: Image fusion method comparison results (%) of 200-way zero-shot retrieval on THINGS-EEG: Top-1 and Top-5 (intra-subject).

Method	Subject 1		Subject 2		Subject 3		Subject 4		Subject 5		Subject 6		Subject 7		Subject 8		Subject 9		Subject 10		Avg	
	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5
Original	67.0	91.5	64.0	94.0	70.0	94.0	45.0	84.0	<u>66.5</u>	88.0	72.0	95.0	59.5	90.5	73.0	95.0	<u>63.0</u>	<u>93.0</u>	66.0	<u>96.0</u>	64.6	92.1
Center_blur	<u>68.0</u>	<u>95.5</u>	<u>69.0</u>	93.5	62.0	90.5	<u>51.0</u>	<u>85.0</u>	62.0	89.0	69.5	<u>95.5</u>	61.5	91.5	<u>74.5</u>	<u>97.5</u>	60.0	90.5	<u>71.0</u>	95.0	<u>64.9</u>	<u>92.4</u>
Saliency_blur	60.0	88.5	54.5	88.0	62.5	92.5	47.5	80.5	51.0	83.5	70.0	90.0	60.5	90.0	67.5	95.0	50.5	80.0	68.5	92.0	59.3	88.0
Linear	66.0	91.0	62.0	93.5	66.0	93.5	47.5	81.5	60.5	84.5	72.5	95.5	65.5	<u>95.5</u>	68.5	96.5	60.0	91.0	63.0	92.5	63.2	91.5
Learnable_mask	66.0	93.0	62.0	<u>94.0</u>	70.0	95.0	46.0	82.0	61.5	87.0	74.5	95.0	65.0	95.0	68.0	94.0	61.5	92.0	66.5	93.5	64.1	92.1
Attention (Ours)	74.0	98.0	80.0	97.5	69.0	94.0	51.0	92.5	67.5	94.0	67.5	96.5	63.0	95.5	83.0	99.0	69.0	96.5	79.0	97.5	70.3	96.1

correctly matched EEG–image pairs and red points denote incorrect matches. The majority of correct pairs cluster tightly, further validating the robustness of the cross-modal alignment.

fig. 10 visualizes the learned frequency-band selection weights under the subject-dependent setting, where the Beta band consistently receives the highest weight, aligning with its established role in visual attention.

figs. 11 and 12 examine the effect of information-guided frequency-band screening in a subject-independent setting. Compared with the subject-dependent results, the band-selection weights become more concentrated in the Beta band, reflecting its consistent role in sustained visual attention. In the time domain, the amplitude range is reduced from about $[-1, 1]$ to $[-0.6, 0.8]$, indicating a moderated denoising effect. In the frequency domain, components above 50 Hz are suppressed, while the 20–40 Hz range is notably enhanced. The resulting EEG topography shows activation not only over the occipital region but also extends toward the right-parietal area; this right-parietal extension likely reflects enhanced processing and integration of non-central visual information to compensate for inter-subject variations in gaze patterns and attentional trajectories. The overall denoising performance is less pronounced than in the subject-dependent case, which can be attributed to higher inter-individual variability in both noise characteristics and cognitive strategies.



Figure 7: Successful retrieval examples.

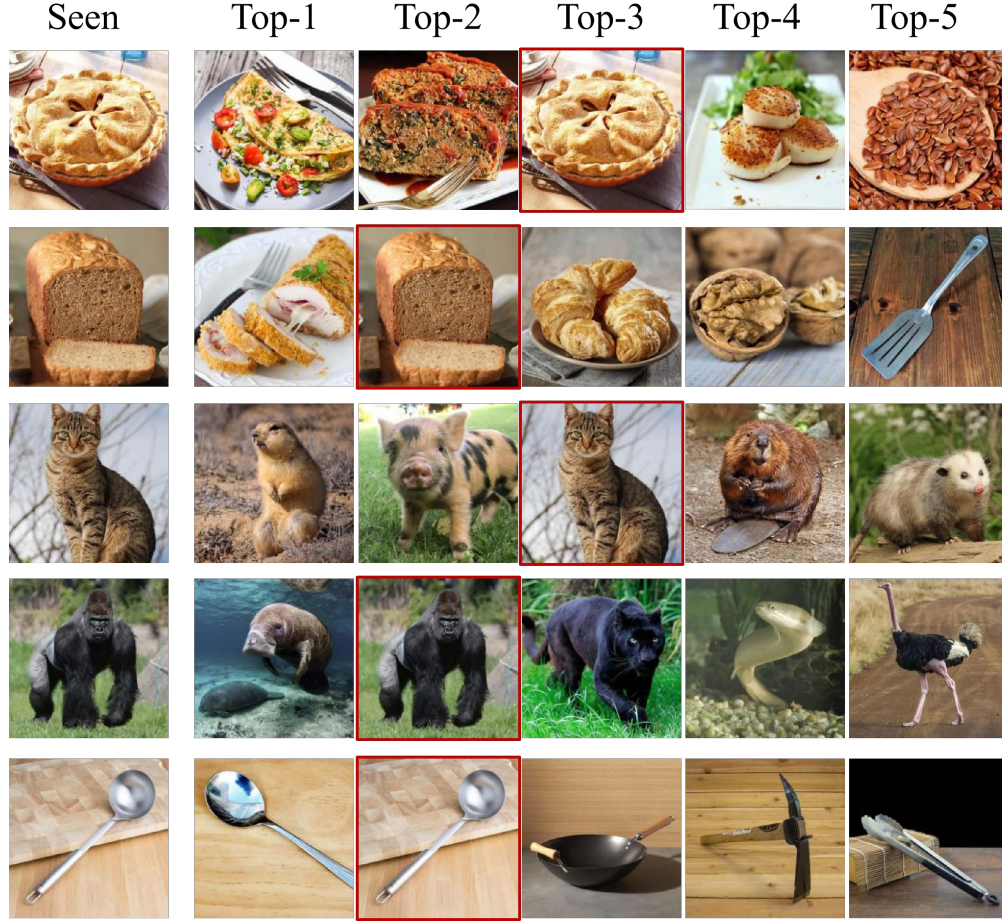


Figure 8: Failed retrieval examples.

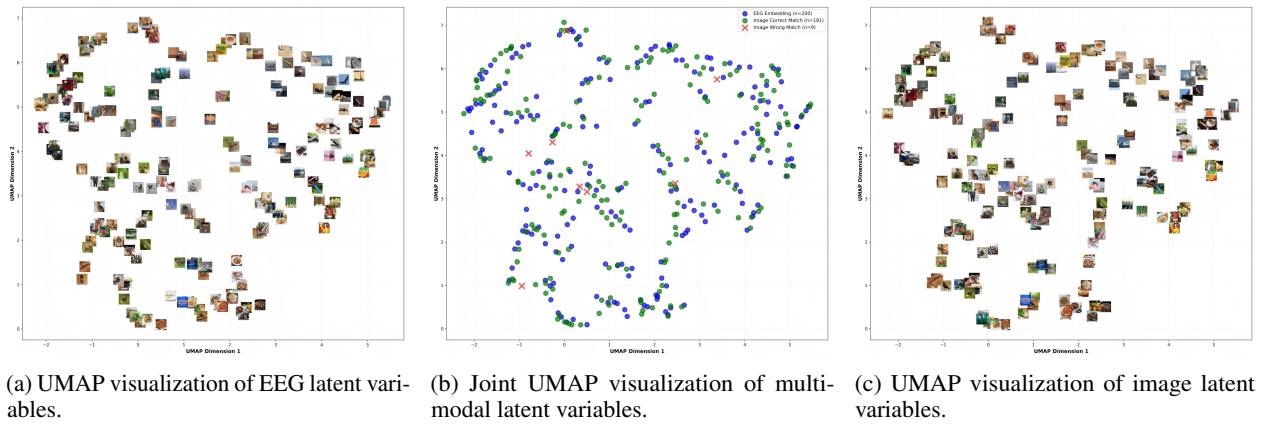


Figure 9: UMAP dimensionality reduction visualization. (b) green represents correct matches, red represents incorrect matches.

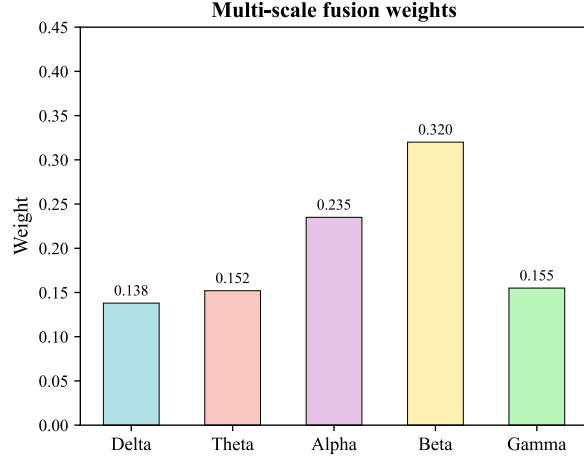


Figure 10: Frequency band selection weights in subject-dependent.

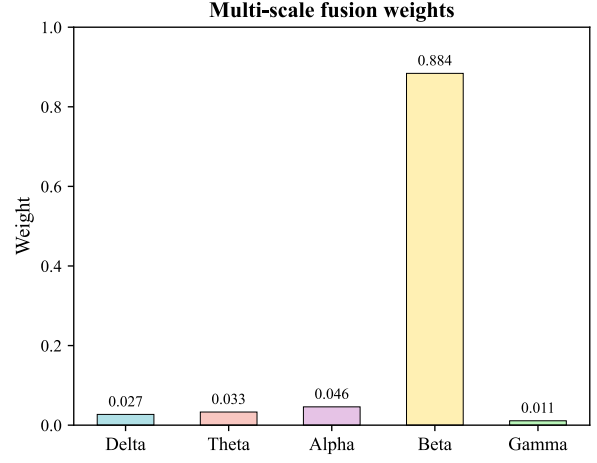


Figure 11: Frequency band selection weights in subject-independent.

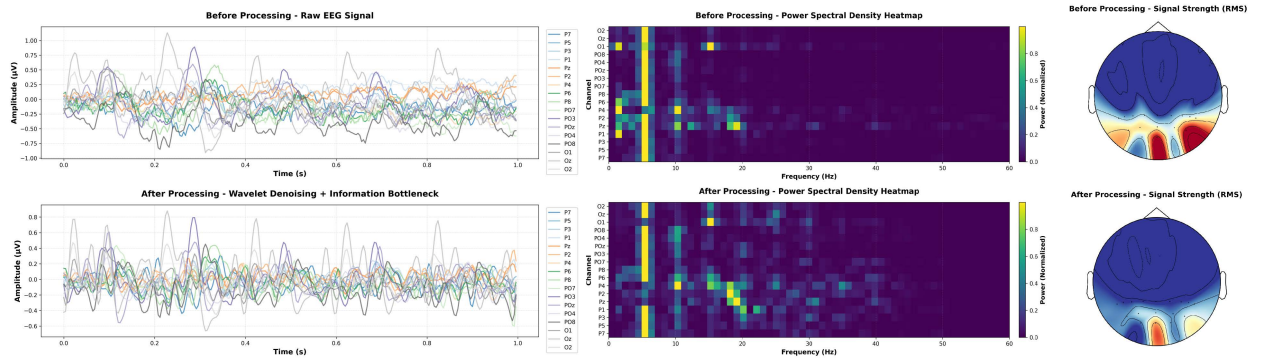


Figure 12: Visualization of EEG signals before and after information-guided frequency band screening under subject-independent setting in the time domain (left), frequency domain (middle), and EEG topographic map (right).